Requirements & Tech Stack

Overview

This document captures the full functional requirements for the Animal Adoption Management System, organized by user role. Requirements are tagged with a priority (Must Have / Should Have / Nice to Have) and a scope flag (In Scope / Future Scope / Out of Scope) based on the initial requirements gathering session. It also defines the tech stack for the development of the system.

1. Functional Requirements

1.1 Adopter

ID	Requirement	Priority	Scope
A-01	Users can create and update their profile with personal information including name, email, phone, address, age, and profile photo, identity & verification (Government ID type + number (for identity verification before adoption approval), ID verification status), household information (Housing type (apartment, house, condo), whether they own or rent, landlord contact / pet policy confirmation (critical — many adoptions fall through here), number of people in household, number and ages of children in household, lifestyle & compatibility (Employment status / work-from-home status, Activity level (sedentary, moderate, active), Yard availability (yes/no), Experience with pets (first-time owner, experienced, professional – pre-qualify), Current or past pets (species, breed, status), Adoption Preferences (Preferred species / breed, Preferred age range of pet, Preferred size, Open to special needs animals (yes/no)), Account & System Fields (Account creation date, Last login date, Account status (active, suspended, banned), Email verification status), Approval status, Adopter type - Fosterer/Owner	Must Have	In Scope
A-02	Users can search for animals using filters: species, breed, age, location, and compatibility with children or other pets.	Must Have	In Scope
A-03	Search results are scoped across all branches of the organisation (multi-shelter network).	Must Have	In Scope
A-04	Users can save animals to a favorites list and remove them at any time.	Must Have	In Scope
A-05	Users can submit an adoption application for a specific animal.	Must Have	In Scope

ID	Requirement	Priority	Scope
A-06	Users can track the status of their application (Pending, Approved, Declined, Withdraw).	Must Have	In Scope
A-07	Users can schedule a shelter visit or virtual meet-and-greet with an animal.	Must Have	In Scope
A-08	Users can access pet care and training resource guides.	Should Have	Out of Scope
A-09	Automated reminders for vaccinations, vet visits, and grooming tasks post-adoption, where they want to be notified (email/SMS).	Must Have	In Scope
A-10	Users can share adoption success stories and photos.	Nice to Have	Out of Scope
A-11	AI adopter behavior rater – Based on adopter history, flags accounts who regularly return pets or don't attend vet appointments	Nice to Have	Out of Scope
A-12	Adopter History	Must Have	In Scope
A-13	Submitting an adoption application requires a non-refundable processing fee, paid via Stripe Checkout before the application record is created.	Must Have	In Scope
A-14	AI compatibility matcher: adopters take a short personality quiz and see their top matching available pets, each with a short 'why this match' reason.	Must Have	In Scope

1.2 Shelter Staff

ID	Requirement	Priority	Scope
S-01	Staff can approve and deactivate staff profile	Must Have	Must Have
S-02	Staff can add, update, and manage animal profiles including photos, species, breed, age, weight, height, blood group, behavior/personality, and colour, Microchip number (unique identifier, industry standard), Intake date, intake type (stray, surrendered, transferred from another shelter), adoption status (incoming, available, pending, adopted, fostered, transferred, deceased), shelterID	Must Have	In Scope
S-03	If a shelter is at full capacity, staff must assign the incoming animal to another shelter branch.	Must Have	In Scope
S-04	Staff can review, approve, or decline adoption applications.	Must Have	In Scope
S-05	Staff can communicate with applicants directly through the system.	Should Have	In Scope
S-06	Staff can manage volunteer applications, schedules, and task assignments.	Must Have	In Scope
S-07	Staff can accept and manage donations made to their shelter branch.	Must Have	In Scope

ID	Requirement	Priority	Scope
S-08	Staff can create and manage adoption events, fundraisers, and community programs.	Must Have	In Scope
S-09	Staff can track medical history, vaccinations, treatments, and special needs for each animal.	Must Have	In Scope
S-10	Staff can view reports on adoption rates, animal intake/outflow, and medical treatments.	Should Have	In Scope
S-11	Animal Transfer Management (Initiating a transfer request, Accepting/rejecting a transfer from another branch, Transfer history log)	Must Have	Must Have
S-12	Branch-scoped access: All staff actions are scoped to their assigned shelter branch by default. Cross-branch visibility is read-only (e.g. for transfer purposes).	Must Have	Must Have
S-13	Foster management	Nice to Have	Out of Scope
S-14	AI health passport summary: staff can generate a short summary of the pet's records plus points to review, each citing the record it came from; no diagnoses, medications or dosages.	Should Have	In Scope

1.3 Veterinarian

ID	Requirement	Priority	Scope
V-01	Vets can view and manage upcoming animal health appointments.	Must Have	In Scope
V-02	Vets can record vaccinations including vaccine name, type, status, last administered date, and due date.	Must Have	In Scope
V-03	Vets can access the universal health passport for any animal, even if it has transferred between shelter branches.	Must Have	In Scope
V-04	AI health passport summary: vets can generate a short summary of the pet's records plus points to review, each citing the record it came from; no diagnoses, medications or dosages.	Should Have	In Scope

1.4 Volunteer

ID	Requirement	Priority	Scope
VL-01	Volunteers can view their assigned tasks and current task status.	Must Have	In Scope
VL-02	Volunteers can view the status of their application (Pending, Approved, Rejected).	Must Have	In Scope
VL-03	Volunteer Sign up – Basic information (name, email, phone, address, age, and profile photo, identity & verification (Government ID type + number (for identity verification before adoption approval), ID	Must Have	In Scope

ID	Requirement	Priority	Scope
	verification status)), task preferences, available timings, shelter ID		
VL-04	Volunteers can manage and update their schedule.	Must Have	In Scope
VL-05	Volunteers can mark tasks as complete.	Must Have	In Scope

1.5 Donor

ID	Requirement	Priority	Scope
D-01	Donors can make a monetary donation to a specific shelter branch. (Payment gateway)	Must Have	In Scope
D-02	Donation receipt/confirmation (email)	Must Have	In Scope
D-02	Sign up - Basic information (name, email, phone, address, age, and profile photo	Must Have	In Scope
D-03	Donors can view their full donation history including date, amount, about the impact (TBD) and shelter.	Must Have	In Scope

1.6 Admin

ID	Requirement	Priority	Scope
AD-01	Admin can create and manage shelter branch records including address and capacity.	Must Have	In Scope
AD-02	Admin can assign a staff member as the designated manager of a shelter branch.	Must Have	In Scope
AD-03	Admin has organisation-wide visibility into all shelter operations and transfers.	Must Have	In Scope
AD-04	Admin can view system-wide reports/analytics.	Must Have	In Scope

2. Non-Functional Requirements

ID	Category	Requirement
NF-01	**Security**	All passwords must be hashed using bcrypt. Authentication handled via JWT tokens with appropriate expiry.
NF-02	**Security**	Role-based access control (RBAC) must restrict all API endpoints to authorised roles only.
NF-03	**Performance**	API response times should remain under 500ms for standard CRUD operations under normal load.
NF-04	**Scalability**	The system must support multiple shelter branches without architectural changes — new branches added via admin panel only.

ID	Category	Requirement
NF-05	**Availability**	The application should target 99.5% uptime in production. Availability is monitored by UptimeRobot, which checks the API every 10 minutes and emails an alert on downtime.
NF-06	**Data Integrity**	All adoption-critical data (applications, medical records, transfers) must be persisted with ACID-compliant transactions.
NF-07	**Geolocation**	Location-based animal search must use PostGIS for efficient proximity queries across shelter branches.
NF-08	**Maintainability**	Codebase must include unit tests (Jest), integration tests (Supertest), and end-to-end tests (Cypress) with minimum 70% coverage.
NF-09	**Usability**	The UI must be responsive and accessible on both desktop and mobile browsers.
NF-10	**Data Privacy**	Sensitive fields (government ID, passwords, medical records) must never be exposed in API responses beyond what the requesting role is authorised to see.
NF-11	**API Documentation**	All endpoints must be documented via Swagger/OpenAPI.
NF-12	**Error Handling**	The API must return consistent, structured error responses.
NF-13	**AI Usage**	AI calls send no personal data (no names, contact details or addresses). AI is optional: the application works without it and falls back to rule-based scoring.

3. Tech Stack

The following table outlines the tools and technologies chosen for the system, organized by architectural layer, along with the rationale behind each selection.

Layer	Tool / Technology	Role	Why This Choice
Frontend	**React**	UI Component Library	Component-based architecture enables reusable UI elements across shelter portals and adopter dashboards; large ecosystem supports rapid feature development.
Frontend	**Tailwind CSS**	Utility-First Styling	Eliminates custom CSS overhead; consistent design system with responsive utilities ideal for a multi-role, multi-screen application.
Frontend	**React Router**	Client-Side Routing	Declarative routing for SPA navigation across role-specific views (Admin, Staff, Adopter, Vet) without full-page reloads.
State Management	**Zustand**	Global State	Lightweight and boilerplate-free compared to Redux; ideal for managing auth sessions and cross-component state like favorites and application status.

Layer	Tool / Technology	Role	Why This Choice
State Management	**TanStack Query**	Server State & Caching	Handles async data fetching, caching, and background sync - critical for real-time adoption status and appointment updates.
Backend	**Node.js + Express**	REST API Server	Non-blocking I/O suits concurrent shelter requests; Express provides a minimal, flexible framework for building the API layer quickly.
Backend	**multer**	File uploads	Parses multipart photo/ID uploads with type and 5 MB size limits.
Backend	**helmet**	HTTP security headers	Sensible secure defaults for every API response.
Backend	**zipcodes**	Offline geocoding	Turns US ZIP codes into coordinates for PostGIS without an external API.
Backend	**node-cron**	Scheduled jobs	Runs nightly maintenance jobs.
ORM	**Prisma**	Database Access Layer	Type-safe queries with auto-generated client reduce runtime errors; schema migrations are version-controlled and easy to maintain.
Database	**PostgreSQL**	Primary Relational DB	ACID compliance and strong relational support handle complex joins across pets, applications, and multi-shelter transfers reliably.
Database	**PostGIS**	Geospatial Extension	Enables location-based animal search and shelter proximity filtering - a core feature for the multi-shelter network.
Database	**Supabase**	DB Host + Media Storage	Managed Postgres hosting with built-in storage for pet photos and documents; reduces DevOps overhead.
Storage	**Supabase Storage**	File storage	Public pet-images and private government-ids buckets; private files served only via short-lived signed URLs.
Auth	**JWT**	Stateless Authentication	Scalable token-based auth supports multiple roles (Admin, Staff, Adopter, Vet) without server-side session storage.
Auth	**bcrypt**	Password Hashing	Industry-standard adaptive hashing algorithm protects user credentials with salted hashes resistant to brute-force attacks.

Layer	Tool / Technology	Role	Why This Choice
Containers	**Docker**	Containerization	Ensures environment parity across development, testing, and production; simplifies onboarding for new contributors.
Version Control	**GitHub**	Source Control	Industry-standard Git hosting with PR workflows, branch protection, and CI/CD integration.
API Docs	**Swagger/OpenAPI**	API Documentation	Auto-generates interactive API docs from annotations - facilitates frontend-backend handoffs and third-party integrations.
Payments	**Stripe**	Payment Processing (Checkout)	Handles the $15 adoption-application fee end-to-end via hosted Checkout + webhook-confirmed payment — card data never touches the app, so PCI scope stays entirely with Stripe.
AI	**Groq (openai/gpt-oss-20b)**	Matcher reasons + passport summary	Free tier; open-weight model; OpenAI-compatible API called with plain fetch from the server (no SDK), so switching provider (e.g. local Ollama) is an env change.
Testing	**Supertest**	HTTP Assertions for integration tests	Drives the real Express app over HTTP inside Jest (register → login → call the actual endpoint), so integration tests exercise the full request/response cycle, not isolated functions.
Testing	**Cypress**	End-to-end browser tests	Drives the real app in an actual browser for flows that only exist client-side (e.g. the post-login redirect-back flow) — Jest/Supertest never render React, so they can't reach these.
Testing	**Jest**	Unit & Integration Tests	Fast, zero-config test runner for backend logic and utility functions; integrates cleanly with Node/Express.
Deployment	**Render**	Backend Deployment	Managed cloud platform that runs the API's Docker image on a free plan with no expiry (the database stays on Supabase) and redeploys from GitHub, with minimal DevOps overhead.
Development	**Vercel**	Frontend Deployment	Zero-configuration deployment for React apps. Connects directly to GitHub and auto-deploys on every push, with a global CDN for fast load times out of the box.

Layer	Tool / Technology	Role	Why This Choice
Monitoring	**UptimeRobot**	Uptime Monitoring	Free monitor that calls the API every 10 minutes, so Render's free plan never sleeps (no 30–60 s cold start) and the Supabase database stays active; emails an alert if the API goes down.
Development	**Cursor**	AI-Powered IDE	AI-powered IDE that understands the entire codebase as context, enabling multi-file edits across React, Express, and Prisma layers in a single instruction. Ideal for a full-stack project with multiple interconnected portals.
Planning	**Trello**	Task & Sprint Board	Visual Kanban-style board for tracking sprint tasks, backlog grooming, and progress across the 8-sprint plan; simple and effective for solo project management.
Design	**Figma**	UI Prototyping	Collaborative design tool for wireframing and high-fidelity prototypes before frontend implementation.
Design	**dbdiagram.io**	ER-Diagram	Dedicated ER diagram creation tool.
Planning	**UXPressia**	User Persona Creation	Dedicated persona creation tool that structures user research into detailed, visually rich profiles — ideal for defining the goals, pain points, and behavioural patterns of the six user types (Admin, Shelter Staff, Adopter, Vet, Volunteer and Donor) before frontend design begins.
Docs	**Word/Google Docs**	Project Documentation	Centralized knowledge base for requirements, sprint planning, API references and development documentation.